

TrES-5b

R-2005

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_150905 · created 2026-08-07T16:25:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457270.8604 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.36 +/- 1.94 [%]
Orbital Inclination (inc)	86.57 +/- 2.93
Airmass coefficient 1 (a1)	1.49 +/- 0.87
Airmass coefficient 2 (a2)	-0.26 +/- 0.32
Scatter in the residuals of the lightcurve fit is	56.65 %
Best Comparison Star	None
Optimal Aperture	4.26
Optimal Annulus	6.29
Transit Duration (day)	0.083 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.053 vs 0.142 (deviation 0.56σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.47σ)
Mid-time O–C	-4.3 min
Depth SNR	2.5
Residual scatter	56.65 %

Light curve

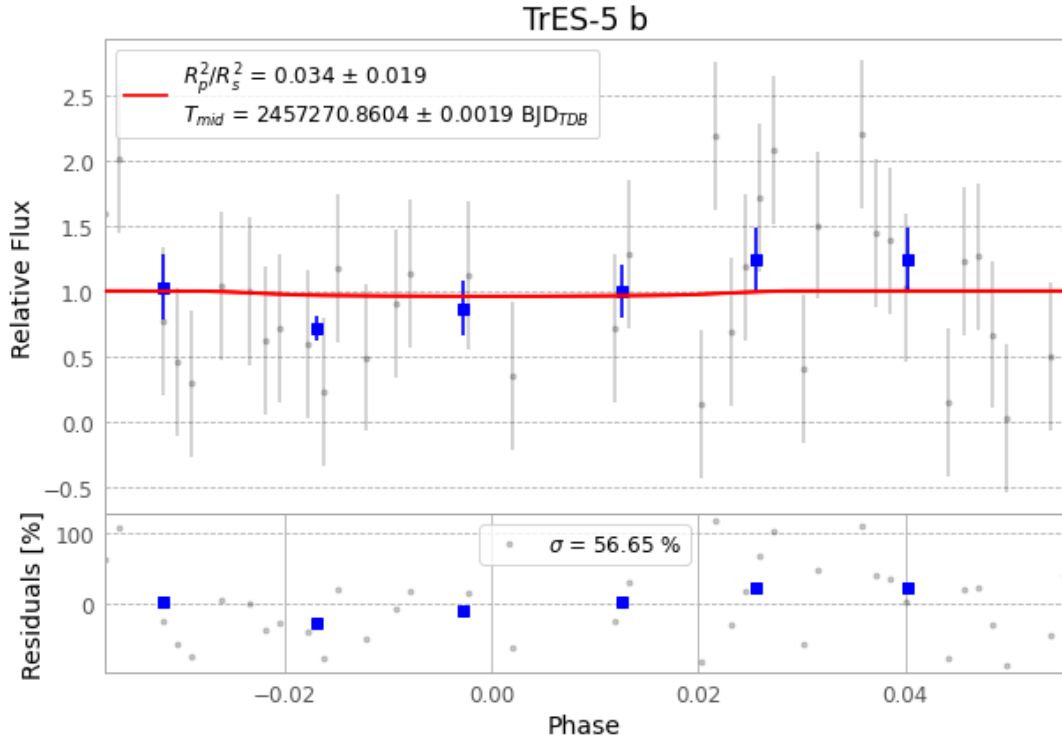


Figure 1 — FinalLightCurve_TrES-5 b_2015-09-05.png (SHA-256 324205b806080de1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e951bc32c7791370...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), TRES-5150905071512.FITS → TRES-5150905103313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150905.log (40261d8874dd...), FinalLightCurve_TrES-5 b_2015-09-05.png (324205b80608...), FinalLightCurve_TrES-5 b_2015-09-05.csv (3d3077a00ff5...)

Compute resources

Wall time 188.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:25Z.